WEEK 11 TA SESSION

$a \iff \alpha,\ b \iff \beta, e \iff \mathcal{E}$

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1. SOFT MARGIN SVM

$$\min_{w,e} \frac{||w||^2}{2} + C\sum_{i=1}^{n} e_i$$

$$st.\ 1 - \left(w^T x_i\right) y_i - e_i \leqslant 0$$
$$-e_i \leqslant 0$$
$$\forall i$$

$$\min_{w,e} \max_{a\geqslant 0, b\geqslant 0} \frac{||w||^2}{2} + C\sum_{i=1}^{n} e_i + \sum_{i=1}^{n} a_i\left(1 - \left(w^T x_i\right) y_i - e_i\right) - \sum_{i=1}^{n} b_i(e_i)$$

$$\max_{a\geqslant 0, b\geqslant 0} \min_{w,e} \frac{||w||^2}{2} + C\sum_{i=1}^{n} e_i + \sum_{i=1}^{n} a_i\left(1 - \left(w^T x_i\right) y_i - e_i\right) - \sum_{i=1}^{n} b_i(e_i)$$

$$w^* = XYa^*$$

$$a_i + b_i = C,\ \forall i$$

Complementary Slackness Conditions

$$a_i\left(1 - \left(w^T x_i\right) y_i - e_i\right) = 0$$
$$b_i(e_i) = 0$$

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2. ACTIVITY 10.6

1) 200
2) C
3) 0
4) ∞

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3. ACTIVITY 11.1

$C \geqslant a \geqslant 0$
$a + b = C$

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4. ACTIVITY 11.2

$\Rightarrow a_i = 0$
$b_i = 0$
$e_i = 0$
$1 - \left(w^T x_i\right) y_i - e_i \leqslant 0$
$1 - \left(w^T x_i\right) y_i \leqslant 0$
$\left(w^T x_i\right) y_i \geqslant 1$

$\Rightarrow a_i = C$
$b_i = 0$
$e_i \geqslant 0$
$1 - \left(w^T x_i\right) y_i - e_i = 0$
$1 - \left(w^T x_i\right) y_i = e_i$
$1 - \left(w^T x_i\right) y_i \geqslant 0$
$\left(w^T x_i\right) y_i \leqslant 1$

$\Rightarrow 0 < a_i < C$
$0 < b_i < C$
$e_i = 0$
$1 - \left(w^T x_i\right) y_i - e_i = 0$
$1 - \left(w^T x_i\right) y_i = 0$
$\left(w^T x_i\right) y_i = 1$

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5. ACTIVITY 11.3

$\Rightarrow \left(w^T x_i\right) y_i < 1$
$e_i > 0$
$b_i = 0$

$a_i = C$

$\Rightarrow \left(w^T x_i\right) y_i = 1$
$e_i = 0$
$C \geqslant b_i \geqslant 0$
$C \geqslant a_i \geqslant 0$

$\Rightarrow \left(w^T x_i\right) \geqslant 1$

$\Rightarrow a_i > 0$
$0 \leqslant b_i < C$
$e_i \geqslant 0$
$\left(1 - \left(w^T x_i\right) y_i - e_i\right) = 0$
$1 - \left(w^T x_i\right) y_i = e_i$
$1 - \left(w^T x_i\right) y_i \geqslant 0$
$\left(w^T x_i\right) y_i \leqslant 1$

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6. ACTIVITY 11.4

S1 $P(\lambda)$
S2 $P(\lambda)$
...
$\lambda_1, \lambda_2, \lambda_3, \ldots, \lambda_n$

$$E[\widehat{\lambda}] = \sum_{i=1}^{n} \lambda_i = \lambda \rightarrow unbiased$$

$$\sum_{i=1}^{n} \left(\lambda_i - \sum_{i=1}^{n} \lambda_i\right)^2$$

$\lambda = 3$
$2.9, 2.7, 3.1, 3.3, 3.5, 3.1, 2.9, 2.8$

$10, 0, 3, 5, 11, 2, 1, 1, 8$

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$100, 1$

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7. ACTIVITY 11.5

$x_1, \ldots, x_5$

bag 1

$x_3 \to 40$

$$1 - \left(1 - \frac{1}{n}\right)^n$$

$f_1, \ldots, f_{10}$

$f_1 \to 0.8$

$f_2 \to 0.4$